

Construction of Communication Information Encryption Transmission System Based on Bayesian Algorithm

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Abstract—Communication not only brings convenience to people, but also produces information. Malicious intrusion programs continue to threaten the network security in the network, resulting in a great threat to the privacy of users and becoming a potential safety hazard. In order to protect the security of network communication information, it is necessary to encrypt the key of communication message information. Naive Bayesian classification is an effective machine learning algorithm, which has important applications in text detection and medical diagnosis. In order to improve the security and real-time transmission of network communication information, this paper proposes a real-time encrypted transmission system of network communication information based on Bayesian algorithm, which can efficiently realize safe and naive Bayesian training and classification on encrypted data. The simulation results show that with the increase of the quantity of experiments, the accuracy of this algorithm is stable at about 95%, and the real-time wavelength tends to be stable, which is 18.85% higher than the traditional algorithm on average. The proposed scheme not only protects the privacy of outsourced data sets, naive Bayes model, samples to be classified and classification results, but also effectively reduces the computation and communication overhead. This method improves the security of the network and provides an effective guarantee for the security and real-time performance of the wireless communication network.

Keywords—Bayesian algorithm, Communication encryption, Information transmission, Information encryption

I. INTRODUCTION

With the rapid development of the Internet, network communication has been widely used in video conferencing, information transmission and other fields, bringing great convenience to people. But while network communication brings convenience to people, it also produces information. Malicious intrusion programs constantly threaten the network security in the network, resulting in a great threat to the privacy of users, which has become a potential safety hazard[1]. It can be seen that the problem of network security has quickly become a major issue related to human life and survival. In large-scale wireless ad hoc networks, nodes need to communicate with a large quantity of internal nodes. In order to ensure the security of communication between nodes, before sending information, nodes must encrypt the information with shared key, which plays a vital role in the security of the whole network[2]. Unlike ordinary networks, wireless sensor network nodes are limited in computing power, energy and communication bandwidth, so they require high passwords[3]. At present, the key pre-distribution algorithm and random key pre-distribution

algorithm are widely used in the security management algorithms of wireless sensor networks. In the current mainstream wireless network key distribution management methods, in order to ensure the security of communication, the keys of nodes should be updated regularly. The encryption algorithm technology to ensure the security of network communication information has been well developed and paid more and more attention by researchers, which is a hot issue in the research of network information security[4].

Authentication and encryption of wireless sensors are important components of the security mechanism of wireless sensor networks, and the security of key is the foundation of all wireless sensor security. The nodes in the network are randomly distributed according to the field situation, which relies on the cluster head nodes between regions as a link, and the keys among the nodes are updated one by one through mutual communication[5]. In some sites, the distribution of nodes is sparse and the distance between nodes is large. In the process of updating the key of nodes, if the distance between nodes is large, the speed of updating the key will decrease, and the key of some nodes will not be updated for a long time[6]. Naive Bayes classification is an effective machine learning algorithm, which has important applications in text detection and medical diagnosis[7]. In order to protect the security of network communication information, it is necessary to encrypt the key of communication message information. However, when the communication is attacked by the network, some communication messages are intercepted and deciphered, and the traditional AES encryption algorithm with the same key is used for encryption and decryption, which cannot effectively prevent the interception and deciphering of the message by the network attack, resulting in the problem of low network communication security[8]. In order to improve the security and real-time transmission of network communication information, this paper proposes a real-time encrypted transmission system of network communication information based on Bayesian algorithm, which can efficiently realize safe and naive Bayesian training and classification on encrypted data.

II. DESIGN RESEARCH

A. Communication System Theory

Wireless sensor network nodes have limited energy, computing power and communication ability. Usually, sensor nodes are randomly distributed in a certain area, so the performance of sensor network can be improved by

reasonable deployment of node distribution. The conventional method of wireless network node encryption communication is to update the key of adjacent nodes according to the key standard of the nodes themselves. If the key update of one node is completed, the communication connection between nodes is established and the next node is notified to update the key[9]. By reasonably dividing the whole wireless sensor network, some nodes can have a

neighborhood relationship. This neighborhood relationship is the basis of key establishment, and the key pool is divided uniformly, and the division is consistent with the node area division, which can further improve the probability of sharing keys. Some nodes correspond to multiple keys, which is called cross-correspondence. The communication mechanism of wireless network is shown in Figure 1.

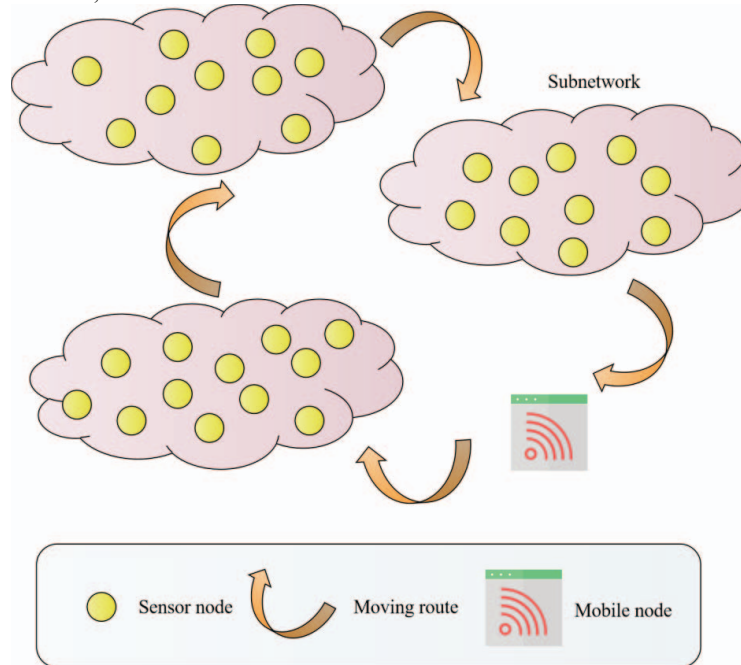


Figure 1 Wireless communication network communication mechanism

B. Problems in Communication System

With the continuous development of wireless communication network, the life style of human beings has changed greatly, and the demand for communication data services has increased dramatically, which has also promoted the reform of mobile communication system. At present, the commercial deployment of 5G in the world shows that the wireless communication network has made breakthrough progress, and officially opened the curtain of the Internet of Everything. If the user's data requires higher security, the highest level of network encryption technology should be adopted. Before the data in the data packet is sent to the local area network, it should be encrypted by software or hardware. Only those sites with correct keys can recover and read these data[10]. Wireless LAN has some other good security. In order to achieve wider coverage, the satellite network will be introduced to create an air-space integrated network, so as to realize the three-dimensional and global ubiquitous coverage of the wireless communication network.

Whether there is a wireless network segment or not, most LANs must have a certain level of security measures. Even the wired network is vulnerable to internal curiosity, external invasion and wiretapping. No one wants to risk exposing the data on the LAN to uninvited guests and malicious intrusions. In order to make the application more comprehensive, based on the introduction of artificial intelligence, big data and other technologies, a twin virtual

digital world is generated according to the physical world, through which the information of the physical world can be transmitted. This trend also puts forward higher requirements for privacy protection and security of wireless communication networks in the future.

C. Bayesian Network Theory

Technically speaking, Bayesian network is a language that systematically describes the relationship between random variables. Theoretically, only a joint probability distribution is needed to make inferential calculations between random variables. Bayesian network was originally developed to deal with uncertainties in the field of artificial intelligence[11]. Its purpose is to introduce knowledge and experience into the computer system, so as to establish an expert system with the ability to intelligently deal with indefinite problems. Probability theory, as a mathematics to study randomness, has strong logicity and explanatory power to data conclusions when dealing with uncertain problems. As the quantity of variables increases, the complexity of joint probability distribution increases exponentially[12].

III. EXPERIMENTAL RESEARCH

A. Bayesian Theoretical Model of Data Processing

Traditional key distribution algorithms are all based on updating the key of the network one by one. In the current wireless network, the distribution of nodes has strong randomness, and some nodes are at a late distance, which

will cause a long time to be consumed in the process of notifying the key update. If the network nodes are distributed loosely and the hopping distance between nodes is far away, it is likely that the keys of some nodes will not be updated for a long time. Using the conditional independent relationship between random variables can reduce the complexity of joint distribution expression in computer system, and improve the speed of inferring the edge distribution of given conditions in joint distribution, so that the probability method can be successfully applied to large-scale problems. Let the probability distribution of random variable set $X = \{X_1, X_2, \dots, X_n\}$ be $P(X_1, X_2, \dots, X_n)$. If all variables are $P(X_1, X_2, \dots, X_n)$, $2^n - 1$ parameters are needed to determine the joint distribution. And through Bayesian formula, the joint distribution can be written as:

$$\begin{aligned} P(X_1, X_2, \dots, X_n) \\ = P(X_1)P(X_2|X_1)\dots P(X_n|X_1, X_2, \dots, X_{n-1}) \\ = \prod_{i=1}^n P(X_i|X_1, X_2, \dots, X_{i-1}) \end{aligned} \quad (1)$$

For $\forall X_i \in X$, if $\pi(X_i) \subseteq \{X_1, X_2, \dots, X_{i-1}\}$

exists, the conditions of X_i assimilation $\{X_1, X_2, \dots, X_{i-1}\} / \pi(X_i)$ are independent at a given $\pi(X_i)$, and the above formula can be changed to:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i | \pi(X_i)) \quad (2)$$

In Bayesian networks, there are three independent relationships: conditional independence, environmental independence and causal independence. If conditional independence is the independence of Bayesian network topology, then environmental independence and causal independence are the independence of Bayesian network parameters.

B. Security Transmission Model of Wireless Communication Information

A digital encryption system includes two processes: encryption and decryption, including encryption algorithm, ciphertext, plaintext and key. Encryption algorithm is the core part of building a key system. It changes the data structure of the original information into ciphertext, and finally reconstructs the original information through decryption algorithm, as shown in Figure 2.

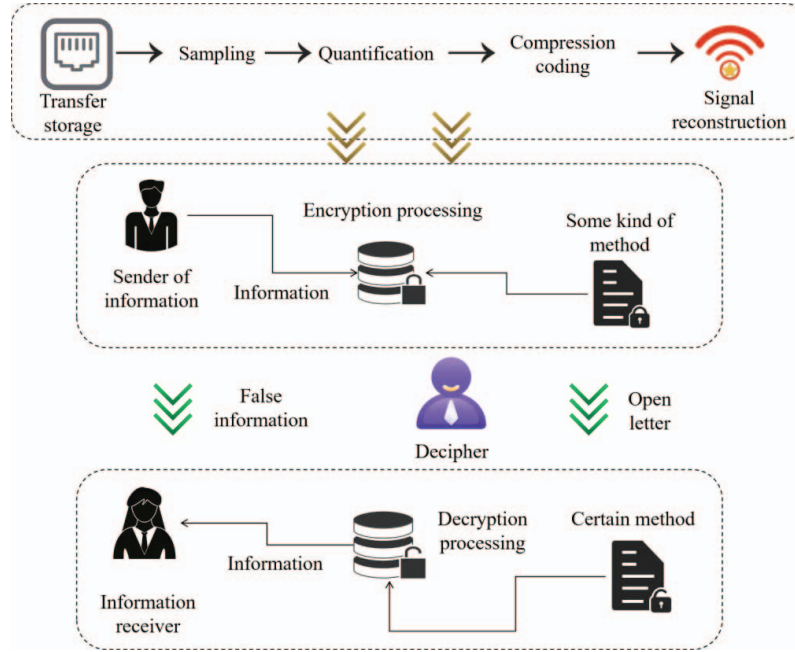


Figure 2 Digital encryption process

X signal is a one-dimensional finite discrete-time signal, which can be expressed linearly by $N \times 1$ -dimensional basis vector $\{\psi_i\}_{i=1}^N$ and its linear combination. Θ is the column vector of $N \times 1$ composed of projection coefficients:

$$\Theta = [\theta_i] = [\langle X, \psi_i \rangle] \quad (3)$$

If the basis vector $\{\psi_i\}_{i=1}^N$ is taken as a column vector, there will be a $N \times N$ -dimensional basis matrix:

$$\psi = [\psi_1, \psi_2, \dots, \psi_N] \quad (4)$$

Then the time domain signal can be expressed as:

$$X = \psi \Theta \quad (5)$$

The X signal and ψ domain signal Θ are different representations of the same signal, and ψ is the transformation base of the signal X . If the signal X is sparse or compressible, the observation signal can be obtained by linear transformation of the observation vector irrelevant to ψ .

IV. EXPERIMENTAL ANALYSIS

The rapid development of network technology not only brings us convenient communication conditions, but also makes our communication information in a dangerous situation that is easily stolen and utilized. Wireless communication network optimization must follow the premise of security, stability and reliability, and ensure the continuous and uninterrupted operation of wireless communication network. Wireless communication network optimization needs to follow the principle of flexibility, adapt to the change of optical transmission technology in time, and ensure the flexibility of information transmission and communication. Figure 3 shows the comparison of simulation running time of different algorithms.

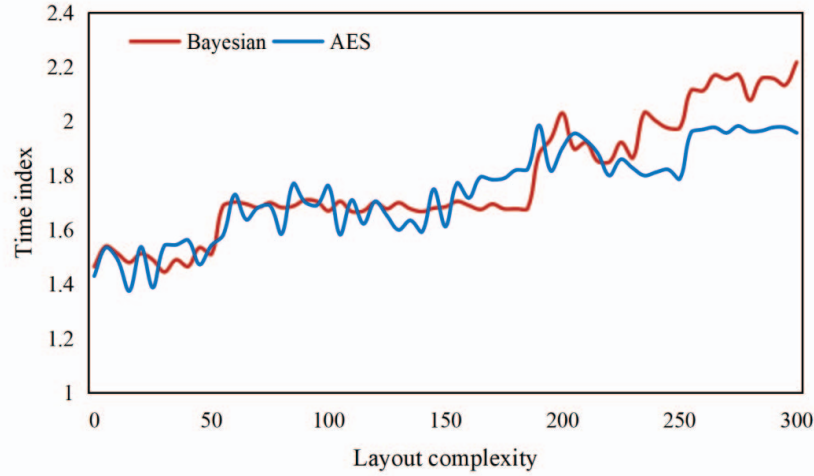


Figure 3 Comparison of simulation running time of different algorithms

It can be seen that although Bayesian algorithm didn't show obvious advantages at the initial stage, when the

complexity of building information is constantly increasing, Bayesian algorithm shows high operating efficiency. The convergence comparison results between AES method and Bayesian algorithm are shown in Figure 4.

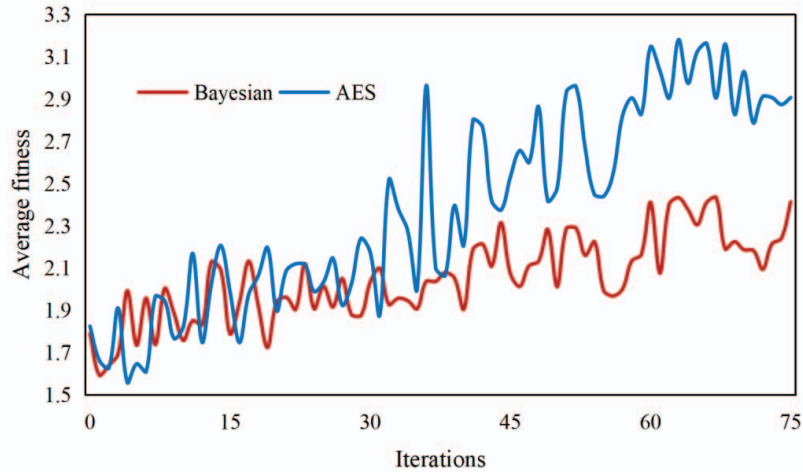


Figure 4 Convergence comparison results

In the communication information encryption stage, firstly, the information to be sent by the sender is encrypted once by its set private parameters, and then additional information is added to the front of the information ciphertext and the system parameters are used for the

second encryption to obtain the ciphertext of the whole data. After the application range of wireless communication network is gradually expanded, the amount of communication information fed back by users is huge. If only the original wireless communication system is used for

processing, it will not only fail to solve the problem in time, but also cause the system to crash. Fig. 5 shows the

comparison of the accuracy of wireless communication network structure optimization algorithms.

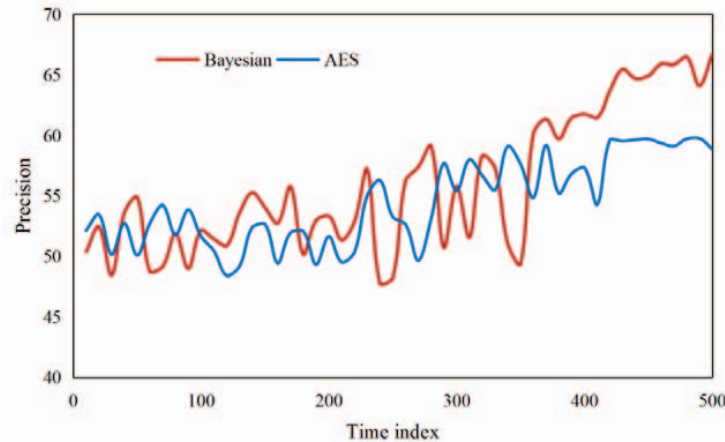


Figure 5 Accuracy comparison of algorithm

It can be seen that with the increase of the quantity of experiments, the accuracy of this algorithm is stable at about 95%, and the real-time wavelength tends to be stable, which is 18.85% higher than the traditional algorithm on average. Mass storage data in wireless communication network can realize collaborative integration of power grid and large-scale clustered data processing. In order to obtain a reasonable cross-transform coefficient, it is necessary to change the position movement status according to the actual situation. If the change range is relatively large, the probability of obtaining new encrypted data can be increased, so as to obtain the most reasonable way to resist network attacks and ensure network security.

V. CONCLUSIONS

In large-scale wireless ad hoc networks, nodes need to communicate with a large quantity of internal nodes. In order to ensure the security of communication between nodes, before sending information, nodes must encrypt the information with shared key, which plays a vital role in the security of the whole network. In this paper, a communication information encryption transmission system based on Bayesian algorithm is proposed. According to the reasonable data length required for encryption, multiple paths between the starting point and the ending point are selected to encrypt the data, thus avoiding sharing a key for encryption and decryption. In this way, after the network attack intercepts the message, even if some messages are deciphered, other messages cannot be deciphered quickly, that is, all messages cannot be completely deciphered. The algorithm uses the deployment and location information of nodes to divide the node information into small areas, and the nodes in the small areas share the key, which overcomes the disadvantages of relying on the key communication of cluster head nodes and ensures the security of the key. The proposed scheme not only protects the privacy of outsourced data sets, naive Bayes model, samples to be classified and classification results, but also effectively reduces the computation and communication overhead. This method improves the security of the network and provides an effective guarantee for the security and real-time performance of the wireless communication network.

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